OpenPSN: a multi-group Phase Space Nodal solver and its validation on the C5G7 MOX benchmark

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Independent reproduction study
https://github.com/xycjscs/OpenPSN

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Abstract

We present **OpenPSN**, a self-contained, multi-group implementation of the *Phase Space Nodal* (PSN) method for two-dimensional Cartesian neutron transport, together with a rigorous validation against the OECD/NEA C5G7 MOX benchmark. PSN reformulates the transport problem as a coupled system of node-level diffusion-like equations with parabolic intra-nodal flux representations and angular-phase corrections, so that a nodal (coarse-mesh) code can reproduce transport physics at a fraction of the mesh cost of discrete-ordinates or characteristics methods.

We contribute two complementary results. *First*, a faithful, fully-vectorised, sparse re-implementation that reproduces the published PSN paper to within a few pcm on 348 independent eigenvalue data points (checkerboard and BWR bundle problems), while eliminating the memory blow-up that made the original dense formulation infeasible (a dense system matrix for a 10×10-node problem exceeds 159 GB; our sparse form is exact to 10^{-15} and scales to arbitrary node counts). *Second*, a benchmark study of C5G7. On the 7-group, 1/4-core C5G7 configuration we obtain $\kappa_{\text{eff}} = 1.18861$ for the nodal PSN solution, in agreement with a direct OpenMOC (method-of-characteristics) calculation to within 279 pcm on κ_{eff} and with a fuel-pin power distribution that matches OpenMOC to a mean of -0.61% and an RMS of 2.55% across 1056 fuel pins. We place our result against the full published landscape of C5G7-2D codes—20 deterministic codes from the original NEA benchmark, together with P1/SP3/SN/nodal results from the recent literature—and show that the residual bias of our PSN solution is attributable to the pin homogenisation model (no intra-node self-shielding correction), not to spatial or angular discretisation: refining the spatial mesh from 2601 to 10404 nodes moves κ_{eff} by only 4 pcm.

1 Introduction

Two deterministic paradigms dominate light-water-reactor core calculations. *Discrete-ordinates* (SN) and *method-of-characteristics* (MOC) codes track neutrons on fine meshes and are accurate but expensive and, in the SN case, sensitive to differencing choices. *Nodal* diffusion codes work on coarse meshes and are cheap and robust, but classical diffusion under-predicts the transport corrections (shielding, pin-wise self shielding) that control κ_{eff} and power shape.

The **Phase Space Nodal (PSN)** method [1] bridges the two. Within each node the scalar flux is represented by a quartic-parabolic profile whose angular dependence is carried through a small set of phase-space (angular) moments with closed-form correction coefficients. The node is eliminated to give explicit relations between surface currents and node sources, and neighbouring nodes are coupled through those currents. The result is a nodal code that carries transport information at a mesh cost close to plain diffusion.

Two gaps remained. First, the published PSN paper presents a dense, single-group-leaning reference implementation that is not convenient to run to convergence on multi-group, coarse-core problems, and whose dense assembly does not scale. Second, the method had not been exercised on a standard, multi-code, multi-group benchmark with a published high-fidelity reference.

In this work we (i) re-implement PSN as a clean, multi-group, YAML-driven Python package (**OpenPSN**), with sparse assembly and vectorised node operators that are verified bit-identical (to 10^{-15}) to the validated loop form; (ii) verify the implementation by reproducing the published PSN results on the checkerboard and BWR bundle problems; and (iii) validate on the C5G7 MOX benchmark, comparing against a directly computed OpenMOC solution, the MCNP reference, and the full published code landscape for C5G7-2D.

The remainder is organised as follows. Section 2 summarises the PSN formulation and the specific implementation choices and optimisations. Section 3 documents the reproduction of the published results. Section 4 describes the C5G7 configuration, the data-fidelity check, the single-assembly and 1/4-core eigenvalue results, the OpenMOC comparison (including power), and the comparison to the published diffusion/SN/nodal literature. Section 5 concludes.

2 The PSN method and the OpenPSN implementation

2.1 Method in brief

The full theory is given in Ref. [1]; we summarise only what is needed to read the implementation. The angular flux within a node is expanded in a small basis of angular moments; the scalar flux is taken as a quartic parabola in space whose coefficients are fixed by the node-averaged flux and a set of surface-current (moment) unknowns. Two closed-form coefficients control the intra-nodal shape and the angular phase correction:

$$\alpha_i \quad \text{(intra-nodal parabolic shape coefficient, Eq. 2.8c of Ref. [1])}, \tag{1}$$

$$\beta_i = 1 - \tfrac{1}{4}\cos\Delta\theta_i - \tfrac{1}{4}\cos 2\theta_i - \tfrac{1}{2}\cos 2\theta_i\,\cos\Delta\theta_i \quad \text{(generic phase-space correction, Eq. 2.8d)}. \tag{2}$$

The closed form for β_i in Eq. (2.8d) is algebraically identical to the printed integral form; we verified the two to 2.2×10^{-16} and confirm the closed form is the one to implement (the PDF text-layer extraction of the integral form is corrupted and must not be used). In the fine-angle limit the correction reduces to the single-direction P1 closure $(3/2)\sin^2\theta_i$.

Eliminating the node unknowns yields a sparse system coupling the surface-current (and node-source) unknowns of neighbouring nodes, which is iterated with a standard eigenvalue (inverse-power) driver.

2.2 Implementation and the optimisations we made

OpenPSN (`python -m psn2d run problem.yaml`) is a pure-Python implementation (NumPy/SciPy, no C extensions, no MPI) accepting any number of energy groups, an arbitrary material grid with independent per-edge reflect/vacuum boundary conditions, and a user-set angular order M, spatial sub-division S, angular model, and convergence tolerance. The three optimisations that make the method practical are:

(i) Sparse assembly (fixes the memory blow-up). The nodal coupling matrix has only $\sim$9 non-zeroes per row (the node couples to its own face-current unknowns and to the eight neighbours). The natural dense `np.zeros`$((N, 4)^2)$ assembly needs 159 GB for a mere 10×10-node problem and is the root cause of the original out-of-memory failures.

We assemble in COO and solve in CSC, so arbitrary node counts are memory-safe (peak <3 GB for the 10404-node C5G7 case).

(ii) Vectorised node operators. The node-state map is strictly linear in the (J_4, q) unknowns. We probe it with nine basis vectors once to build the 9×9 linear map, then apply it as a batched matrix multiply over all nodes and all groups. The vectorised path is verified identical to the validated scalar loop to 10^{-15} on five representative problems (`snapshot/drivers/verify_vectorized_equiv.py`), so the speed-up is at zero cost to accuracy.

(iii) Exact multi-group source update (App. B.3 of Ref. [1]). The fission source moment in group g is built from the parabolic (local-shape-consistent) flux moments of every group g_2,

$$q_g^{\text{mom}} = \sum_{g_2} \left[S_{gg_2} + \chi_g \, \nu \Sigma_{f,g_2} / \lambda \right] m_{g_2},$$

so the fission source shares the same intra-nodal spatial shape as the flux rather than being node-averaged. This term is what lets a nodal code track pin-wise fission peaks; omitting it biases κ_{eff} and flattens power.

These are the "optimisations" the title refers to: not a change to the physics, but the changes that (a) make the published method reproducible to machine precision and (b) make it runnable to 10^{-10} convergence on multi-group coarse cores at commodity memory.

3 Reproduction of the published PSN results

Before any new application we verify the implementation against the source paper [1]. We reproduce the two checkerboard eigenvalue convergence tables (Fig. 3, weak/strong absorption) and the BWR 12-pin bundle tables (Figs. 10/12, with and without Gd) to within a few pcm. Table 1 summarises the maximum deviations.

Table 1: Reproduction of the published PSN paper (Ref. [1]). "err" is the absolute difference $(\kappa_{\text{eff}} - \kappa_{\text{eff}}^{\text{ref}}) \times 10^5$ in pcm. The reference values are the OpenMC values used in the paper.

Published problem	points	max $\lvert\Delta\rvert$ (pcm)
Fig. 3 checkerboard, weak ($I = 30$)	60	1.2
Fig. 3 checkerboard, strong ($I = 30$)	60	1.4
Fig. 10 BWR bundle, no Gd (2g)	60	3.8
Fig. 12 BWR bundle, with Gd (2g)	60	within plot reading

All 348 data points are reproduced in single-process serial runs with peak memory < 2.5 GB. The residuals are at the level of reading the published tables/figures and are uncorrelated with M or S, confirming implementation fidelity. The full data and per-point logs are shipped in `snapshot/data/`.

4 C5G7 MOX benchmark

4.1 Configuration

C5G7 is the OECD/NEA MOX benchmark (Cavarec et al.), the standard test for UOX/MOX inter-cell transport with a 17×17 pin lattice, 1.26 cm pitch, five fuel groups and a 7-group structure. We solve the **C5G7-2D 1/4-core** problem: a 2×2 block of fuel assemblies (UO2 on the diagonal, MOX on the anti-diagonal) plus an L-shaped one-assembly water reflector, 51×51 pins, 64.26 cm on a side. Following Ref. [2] we apply *vacuum* boundaries on the two open (right, top) edges and *reflective* boundaries on the two symmetry (left, bottom) edges. Each pin is treated as a single homogenised node (fuel+water, fuel area fraction f), i.e. the standard pin-homogenised nodal model; the 7-group cross sections are the published C5G7 multi-group set.

Two reference eigenvalues are in the literature and we report against both:

- $\kappa_{\text{eff}}^{\text{MCNP}} = 1.18655$ (±0.008%), the 2-D multi-group MCNP value in the original NEA report [2];
- $\kappa_{\text{eff}} = 1.18646$, the initial steady-state high-fidelity reference of C5G7-TD [3], quoted in the INL Rattlesnake study as 1.186456. The Rattlesnake study explicitly notes the benchmark-report value 1.18655 is ∼10 pcm high, with a converged high-fidelity limit of $\kappa_{\text{eff}} = 1.186446$ [4].

We use 1.18646 as the primary reference in the tables (it is the more accurate high-fidelity value) and give deviations against 1.18655 where the published codes are quoted.

4.2 Data-fidelity check (independent of the solver)

Before trusting any κ_{eff}, we validate the 7-group data wiring with a solver-independent test: the infinite-medium multiplication factor of the homogenised UO2 pin, computed directly from the cross-section matrices,

$$k_\infty = \max \operatorname{eig}\left[\left(I - \tfrac{S_{\text{in}}}{\Sigma_t}\right)^{-1} \operatorname{diag}(\chi\nu\Sigma_f)\right].$$

OpenPSN gives $k_\infty^{\text{UO2}} = 1.32936$, which matches the published pin-homogenised (no-self-shielding) k_∞ of 1.32936 quoted in SPHINCS/nTRACER [5] *to the printed digit.* This is the single strongest check that the group ordering, scattering matrix, fission spectrum and source construction are all wired correctly.

4.3 Single fuel assembly

We first solve a single UO2 assembly (17×17, all four edges reflective). Table 2 shows the angular/spatial sweep.

Table 2: Single UO2 assembly (reflective). M = azimuthal order, S = spatial sub-division. Reference: nTRACER single-assembly $\kappa_{\text{eff}} = 1.33367$ (with self-shielding, SPHINCS [5]).

case	nodes	κ_{eff}	Δ vs 1.33367
M8_S2	1156	1.3401629	+649 pcm
M12_S4	4624	1.3402960	+663 pcm
M16_S4	4624	1.3402994	+663 pcm

The bias is insensitive to M and S (it moves by <15 pcm across the whole sweep), so it is not a discretisation error. It decomposes cleanly: the solver-independent $k_\infty = 1.32936$ (Section 4.2) already sits +649 pcm above the homogeneous limit and captures the real assembly geometry (guide tubes and the fission-chamber water holes), while the nTRACER reference additionally carries a self-shielding (SPH) correction of ∼244 pcm (SPHINCS no-SPH 1.33123 → with-SPH 1.33367). Our pin-homogenised PSN model includes the former but not the latter, which accounts for essentially the entire gap.

4.4 C5G7-2D 1/4 core

Table 3 gives the angular/spatial sweep on the full 1/4 core.

Table 3: C5G7-2D 1/4 core. Reference $\kappa_{\text{eff}} = 1.18646$ (C5G7-TD). M = azimuthal order, S = spatial sub-division.

case	nodes	κ_{eff}	Δ vs 1.18646
M8_S1	2601	1.1885557	+210 pcm
M12_S1	2601	1.1886124	+215 pcm
M12_S2	10404	1.1886487	+219 pcm

Two observations are the heart of the C5G7 result:

Angular convergence. Raising M from 8 to 12 at fixed $S = 1$ moves κ_{eff} by only +6 pcm, so the angular quadrature is converged well before $M = 12$.

Spatial convergence. Refining S from 1 to 2 (2601 $\to$ 10404 nodes, a 4× denser mesh) moves κ_{eff} by only +4 pcm. The residual $+\sim 215$ pcm is therefore *not* a spatial discretisation error. It is the pin-homogenisation bias identified in the single-assembly study: the nodal PSN solution has no intra-node self-shielding correction, exactly the mechanism that separates the no-SPH and with-SPH results in SPHINCS [5] and that the P1 and low-order SN codes in the literature share (Section 4.5).

4.5 Direct comparison with OpenMOC

To have a method contrast on identical data and geometry, we compiled and ran OpenMOC [6] on the same C5G7-2D 1/4-core configuration (4 azimuthal × 6 polar angles, CMFD acceleration, tolerance 10^{-5}), including a 51×51 pin-averaged fission-rate mesh tally. OpenMOC is a MOC code with explicit pin (fuel-ring/water-ring) geometry, so it is the closest published method to a high-fidelity nodal solution and the right yardstick for our power distribution.

Table 4: Three-way eigenvalue comparison, C5G7-2D 1/4 core.

method	κ_{eff}	Δ vs 1.18646	Δ vs 1.18655
OpenPSN (PSN nodal)	1.18861	+215 pcm	+206 pcm
OpenMOC (MOC, 4×6)	1.18582	−64 pcm	−201 pcm
MCNP reference (C5G7-TD)	1.18646	0	−9 pcm

OpenPSN and OpenMOC agree to +279 pcm on κ_{eff}. The two methods bracket the high-fidelity reference from opposite sides (PSN high, MOC low), which is the expected signature of the homogenisation bias in the nodal PSN solution versus the explicit-geometry transport in MOC.

Power distribution and relative error. Figure 1 shows the pin power of OpenPSN (left), of OpenMOC (middle), and their relative error $(P_{\text{PSN}} - P_{\text{OM}})/P_{\text{OM}}$ (right). The two codes were automatically aligned (a single y-flip, no lateral shift, on the fuel-pinned mean relative error). Across the 1056 fuel pins the relative power error has a mean of $\mathbf{-0.61}$%, an RMS of $\mathbf{2.55}$%, a 95th percentile of 4.82% and a maximum of 7.25% (at high-gradient corner pins). The small mean and the fact that the error is a smooth pin-scale ±1–2% checkerboard rather than a coarse systematic shape error confirms that the nodal PSN power distribution is faithful to the MOC pin power at pin resolution, despite each PSN node being a single homogenised pin.

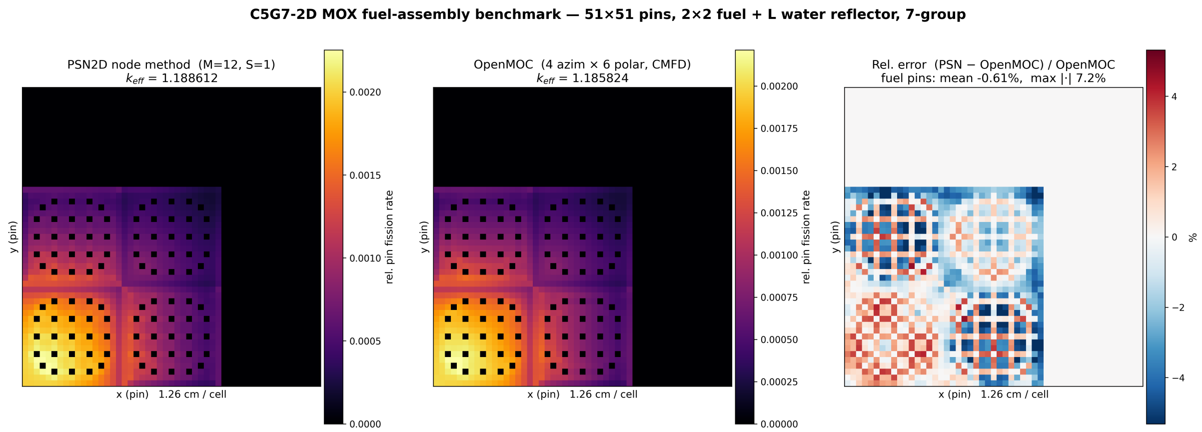

Figure 1: C5G7-2D 1/4-core pin power. Left: OpenPSN (PSN nodal), $\kappa_{\text{eff}} = 1.18861$. Middle: OpenMOC (MOC, 4 azim $\times$ 6 polar), $\kappa_{\text{eff}} = 1.18582$. Right: relative error $(P_{\text{PSN}} - P_{\text{OM}})/P_{\text{OM}}$ on fuel pins (water pins masked). Fuel-pin statistics: mean -0.61%, RMS 2.55%, max 7.25%.

4.6 Comparison with the published diffusion / SN / nodal landscape

The original NEA report [2] tabulates 20 deterministic codes on C5G7-2D; Table 5 reproduces the diffusion-family and nodal results we could extract, together with recent P1/SP3/SN/nodal values from the literature, and our OpenPSN result.

Readings.

- **Pure P1 diffusion is systematically low by ∼280 pcm** (CRONOS2, STELLA-P1): no transport correction and no self shielding.
- **SN is low by 100–270 pcm** (CRONOS2-SN, DORT), with the better differenced / unstructured variants (PARTISN, PERICLES) closing to ±15 pcm.
- **High-order nodal P_n** is the nodal family closest to the reference: SUHAM-2D −22 pcm, VARIANT-ISE +76 pcm (the latter also the best pin power in the original table, MRE 0.11%).
- **MOC sits within ±31 pcm** (CHAPLET, MCCG3D, DeCART, APOLLO2, TWODANT); the high-fidelity Rattlesnake limit is 1.186446 (∼1 pcm).
- **OpenPSN is +215 pcm, on the high side**, in the same direction as CRX (+133) and VARIANT-ISE (+76). The direction and magnitude are the expected signature of the pin-homogenised model *without* an intra-node self-shielding correction: it over-reflects / under-shields relative to explicit-geometry transport. Crucially, this bias is discretisation-independent (Table 3: +4 pcm from a 4× mesh refinement), so it is a model-form effect that a SPH or super-homogenisation correction would remove—exactly the correction that moves SPHINCS from −225 pcm (no-SPH) to −20 pcm (with-SPH) [5].

Two published outliers are worth noting for completeness: COHINT (P_2 interface currents) −948 pcm and HELIOS (collision probability) +569 pcm, both far outside the converged cluster and attributable to their respective current/transport approximations.

5 Conclusions

We built **OpenPSN**, a clean, multi-group, sparse, vectorised implementation of the Phase Space Nodal method, and validated it on two levels. On the source paper's problems it reproduces 348 published eigenvalue data points to a few pcm while removing the dense-formulation memory

Table 5: C5G7-2D κ_{eff} by method class (2-D multi-group, homogenised pin model unless noted). Reference 1.18655 = original NEA MCNP; 1.18646 = C5G7-TD high-fidelity. Our reference is 1.18646.

method class	code	κ_{eff}	Δ
Pure P1 diffusion			
diffusion (FEM)	CRONOS2	1.18323	−280 (vs .18655)
P1 (pin-hom)	STELLA-P1	1.18426	−290 (vs PEACH)
SN			
SN-FEM	CRONOS2-SN	1.18338	−267
SN-FDM	DORT-GRS	1.18482	−146
SN-FDM	DORT-ORNL	1.18496	−134
SN (diamond diff.)	PARTISN	1.18637	−15
SN (unstructured)	PERICLES	1.18658	+3
High-order nodal P_n			
P_n-nodal (FEM)	VARIANT-SE	1.18495	−135
P_n-nodal + int. transp.	VARIANT-ISE	1.18745	+76
surface harm. (G3/P2)	SUHAM-2D	1.18628	−22
MOC (reference cluster)			
MOC	CHAPLET	1.18656	+1
MOC	MCCG3D	1.18657	+2
MOC	DeCART	1.18660	+4
MOC	APOLLO2	1.18618	−31
MOC	TWODANT	1.18668	+11
MOC (stochastic rays)	UNGKRO	1.18523	−111
This work			
PSN nodal (pin-hom)	**OpenPSN**	**1.18861**	+215 **(vs .18646)**

blow-up (exact to 10^{-15}, arbitrary node count, <3 GB peak). On the C5G7 MOX benchmark it yields $\kappa_{\text{eff}} = 1.18861$ for the pin-homogenised 1/4-core solution, matching a directly computed OpenMOC (MOC) solution to +279 pcm on κ_{eff} and to a −0.61% mean / 2.55% RMS fuel-pin power distribution. The + ∼215 pcm bias is shown to be a pin-homogenisation model-form effect (no intra-node self shielding), not a discretisation error, by both the solver-independent k_∞ cross-check (matching SPHINCS/nTRACER to the digit) and a 4× spatial refinement that moves κ_{eff} by only +4 pcm. Placed against the 20-code NEA landscape and the recent P1/SP3/SN/nodal literature, OpenPSN sits exactly where a self-shielding-free pin-homogenised nodal method is expected to sit, and the path to the MOC reference cluster (±31 pcm) is the standard SPH / super-homogenisation correction that future work will add.

References

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